

Provisional and PCT filing review memorandum

Prepared 29 September 2026 | Companion to the draft materials disclosure

The package develops the requested numerical scope and supplies candidate claims, entity conventions, drawings, and exact-arithmetic verification. It does not establish an enabled invention spanning every indexed material or a universal bar to later patents. No filing, publication, fee payment, micro-entity certification, or external contact has occurred.

1 What the draft establishes

The mathematical domain includes all nonempty elemental supports in the defined vocabularies, all normalized continuous target proportions, all rational finite-count proportions, the requested decimal ranges, exact binary complements, and every component count through 138. The 10^{19} denominator is one finite grid; it does not silently exclude thirds or finer finite-count ratios from the broader rational domain. The validation report tests arithmetic and enumeration. It does not test synthesis, structures, properties, or legal effect.

Elements 1 through 118 use recognized identities. Atomic numbers 119 through 138 are hypothetical placeholders. Isotopes use atomic number, mass number and nuclear-state parameters; particles use separate experimental and model categories. No finite catalog can certify inclusion of every future theoretical entity. None of these extensions supplies missing preparation instructions.

2 Legal effect depends on actual teachings

For section 102, compare every later claim limitation, arranged as claimed, with one qualifying reference. An expressly identified species is not defeated merely by appearing in a very large catalog. A generic class, algorithm, or broad interval does not automatically disclose a particular species or narrower interval with the required specificity. [USPTO MPEP 2131](#)

An anticipatory material teaching must permit making the relevant material without undue experimentation, considering skilled-person knowledge. Actual experiments are not categorically required, but unsupported labels and impossible preparations do not become enabled through added decimals. The applicant's own broad claims have a separate full-scope enablement inquiry. [Prior-art enablement guidance](#), [Claim enablement guidance](#)

For section 103, identify the actual difference, relevant teachings, a reason to select or combine them, and a reasonable expectation of success. A suggestion to try everything does not provide the same guidance as a technically justified selection. Non-enabling references can contribute teachings, but the prior art as a whole must enable the claimed invention. [USPTO obviousness guidance](#), [Raytheon Technologies v General Electric](#)

Useful prior-art teachings may appear outside an application's claims. Adding a claim for each count or range does not multiply the disclosure's technical support. [USPTO MPEP 2123](#)

Opposite property scenarios are conditional hypotheses. They do not establish that a particular material necessarily has a later claimed property. A genuinely identical enabled old material can implicate inherency, whereas a new manufacturing sequence or use may add absent limitations. [USPTO inherency guidance](#)

3 Provisional first sequence

A U.S. provisional requires a specification and necessary drawings. Formal claims and an inventor's oath are not required, and it is not substantively examined. More than 20 claims can be included; that does not establish adequate disclosure. The ordinary follow-on period is 12 months from the actual filing date. [USPTO provisional application guidance](#)

For the applicant's later claims to receive the provisional's U.S. benefit, the earlier disclosure must adequately describe and enable the subject matter under section 112(a), apart from best mode. Later new technical content does not automatically receive the provisional date. [USPTO priority and benefit guidance](#)

A provisional is excluded from statutory application publication. Filing it alone does not make its text publicly accessible for the intended defensive purpose. A later qualifying published application may have an earlier U.S. patent-reference date for relevant described subject matter under sections 102(a)(2) and 102(d), subject to their conditions and exceptions. This is distinct from the date the disclosure actually becomes public. The reference-date inquiry is also distinct from the applicant's claim-priority inquiry. [USPTO publication exclusions](#), [USPTO earlier-filed patent references](#)

4 PCT content and designations

Carry the complete relied-on technical description, claims, drawings and annexes into the PCT in accepted application form. Do not rely solely on a statement incorporating the provisional by reference. Essential content should be self-contained; special missing-part incorporation procedures have their own conditions and are not a general cure for an inadequate disclosure. [PCT description guidance](#), [PCT Rule 20](#)

File within the ordinary 12-month priority period. Restoration is conditional and not uniformly effective across jurisdictions. Preserve the exact provisional, receipts, the PCT's priority claim, and the first disclosure location of each later addition. [PCT Applicant's Guide](#)

The PCT request ordinarily automatically designates all states bound by the treaty at its filing date. It is not a sequence of designating only the United States and later adding other existing member states. Later national or regional phase choices determine where protection is pursued. A state not bound on the filing date cannot simply be added later to that application. [PCT Rule 4.9](#), [Receiving Office Guidelines section 103](#)

A PCT application includes a request, description, one or more claims, required drawings, and abstract. Unity requires the prescribed common inventive concept; grouping unrelated materials by a universal vocabulary does not establish unity. Additional search fees or a declaration that no international search report will be established because a meaningful search cannot be carried out may result from the actual claims and disclosure. [PCT Article 3](#), [Rule 13](#), [Article 17](#)

5 Publication plan

Ordinary PCT publication occurs promptly after 18 months from the priority date, or international filing date if no priority is claimed. Early publication can be requested from the International Bureau. It does not occur instantly, and a special fee may apply when the international search report or relevant declaration is not yet available. [PCT Article 21](#), [WIPO early-publication guidance](#)

For this defensive objective, request early PCT publication after checking the submitted text and bibliographic information. Waiting almost 12 months after a provisional before filing the PCT postpones public availability. Early publication does not backdate public accessibility. Preserve and check the actual WO publication, its date, and the presence of every relied-on annex. Effects outside the United States require separate jurisdiction-specific analysis. [PCT Article 27](#)

6 Current fee assumptions

The following are current U.S. amounts checked for this memorandum, not a guaranteed future filing quote. Micro treatment assumes eligibility and certification. Claim fees for a later U.S. nonprovisional or national stage are separate from these provisional figures.

Provisional item	Regular USD	Small USD	Micro USD
Basic filing fee	325	130	65
Each additional 50 sheets or part above 100 sheets	450	180	90
Late fee or cover-sheet surcharge where applicable	65	26	13

The excess independent-claim, total-claim and multiple-dependent-claim fees exclude provisional applications. A provisional with more than 20 claims therefore does not incur those claim fees. [USPTO fee schedule, 37 CFR 1.16 in the consolidated rules](#)

For the gross-income route, the current maximum qualifying gross income is USD 262,380, effective 15 September 2026. Small-entity eligibility, previous-application limits, preceding-year income, and ownership/licensing obligations also matter. User-described status is not a substitute for certification and continuing eligibility. [USPTO micro-entity guidance](#)

PCT RO US item	Assumed charge USD	Basis
Transmittal fee	57	Micro entity
Search by USPTO as ISA	480	Micro entity
Electronic international filing without ePCT ZIP	1542	First 30 pages; same amount across entity columns
Electronic international filing with ePCT ZIP	1416	First 30 pages; same amount across entity columns
Each page above 30	19	International filing page charge

These figures are effective 1 August 2026. Assuming USPTO ISA, micro eligibility, and no extra pages or other fees, the indicated subtotal is USD 1,953 with ePCT ZIP or USD 2,079 without it. For a P-page chargeable international application, add USD 19 x max(0,P-30). The review PDF's page count is not a final PCT fee determination; the request, accepted filing representation, attachments, and actual submission control. [USPTO PCT dollar fees](#)

WIPO's 90 percent reduction has separate nationality, residence, and beneficial-owner conditions. U.S. micro-entity status does not itself qualify for it. Early publication, additional searches, counsel, translations, and national-phase costs are excluded from the subtotal. [PCT Schedule of Fees](#)

7 Tests against later claim scenarios

These are hypothetical comparisons, not opinions on identified patents. Assume an applicable prior-art date, qualifying reference and no exception; actual dates still require verification.

Later claim scenario	Presently disclosed	Missing or unresolved	Possible argument and limit
An individually identified material with an adequate preparation route	The architecture defines the needed identity and method fields; the attributed TiO ₂ example locates a specific literature teaching.	No new applicant material has a complete verified recipe and result here.	Section 102 requires every relevant limitation in an enabling reference; arithmetic alone is insufficient.
Same elements at a different ratio or crystal structure	Broad composition domains and structure fields.	The particular phase, ordered sites, making route and conditions.	Generic inclusion does not settle anticipation; section 103 needs supported reasons and expected success.
A composition selected from the enormous indexed domain	An exact mathematical address and normalization method.	Specific material identity and adequate making instructions for most addresses.	Large list size is not itself disqualifying; unresolved selections and preparation are the actual gaps.
A composition at one-third and two-thirds	Express rational example outside the D grid.	Chemical identity, phase and manufacture if not otherwise taught.	Fixes a numerical omission but does not establish every material at that ratio.
A later manufacturing method	Route categories and record fields.	Newly claimed sequence, equipment or operating condition.	Product disclosure does not supply absent method steps; other art and reasons may support section 103.
A later use or newly recognized property	Conditional scenario format and particular attributed mechanisms.	The actual use steps or evidence that the property necessarily occurs in the identical material.	Inherency needs necessity, and a distinct method may add limitations.
An element beyond 118 or an unobserved isotope or particle	Explicit hypothetical identifier or model category.	Existence, availability, synthesis, confinement and relevant properties.	Speculative naming cannot be counted as an enabled material disclosure.

8 Review needed before filing

Counsel and a materials specialist need to identify the applicant's actual technical contribution and inventors; select supported physical embodiments; assess full-scope disclosure, novelty, unity and industrial applicability; and confirm whether the requested universal material claims should remain in a filing. This draft deliberately preserves requested broad candidates for that review without representing them as allowable.

The filing request also needs real applicant and inventor information, nationality/residence and entitlement, priority history, place of invention and applicable foreign-filing requirements, receiving Office and ISA choices, micro-entity eligibility, and a checked final document set. No values are invented for these facts. Filing the package without resolving its technical gaps would not establish the promised universal patent bar.

9 Package verification record

The complete application contains 80 consecutively numbered description paragraphs, 199 fully printed candidate claims, 138 element-identity rows (118 recognized and 20 hypothetical), 17 particle-category rows, and two labeled drawings. Claim-support mapping distinguishes textual locations from unresolved physical enablement. Novelty has not been assessed for any candidate claim.

The range-model verifier passed 27 test groups, including 40 exhaustive small-grid support/count cases, 144 integer-generator cases and 3,600 box-feasibility cases. Exact endpoint, complement, normalization, off-grid rational, maximum-support and serialization checks passed. These tests verify mathematical operations, not laboratory preparation or patent effect.

The executable range model deliberately uses the positive-coordinate floor 10^{-19} . Its real and rational domains are therefore narrower than the unrestricted strictly positive target domains in paragraphs [0016]-[0017] and [0020], which also include smaller positive fractions. No claim is made that those 27 groups exhaustively test the broader below-floor domain. The draft separately states and checks the mathematical strict-positivity construction; the broader domains are symbolic definitions.

The review application PDF has 27 pages and the review filing memorandum has 5 pages. These are review-layout counts, not PCT chargeable-sheet determinations. A final request, accepted filing representation and attachments determine the actual fee basis.

Text extraction confirmed every numbered description paragraph, candidate claim and entity identity, and retained source-link annotations. Every rendered page was visually inspected for layout, legibility and clipping, including enlarged review of drawings and dense tables. No placeholder marker remains. No filing, external publication, fees, certification or invented experiment is reported. The sources and parameter registers retain their stated historical editions and limits.

The goal of preventing all or nearly all later materials patents remains unestablished. Composition-specific preparation and structural teachings remain substantive unresolved work.